Nalin Singh

To contribute as a Full-Stack Software Engineer, applying skills in React, FastAPI, AWS, and AI-driven Systems to develop high-impact, scalable platforms that enhance user experience and operational efficiency.

Profile Summary

- Possess **nearly 3 years of experience, specializing in full-stack** development, Cloud-Native Applications Architecture, and scalable system Design.
- Currently working as a **Software Engineer at Koch Industries**, leading the development of enterprise-grade workflow and automation platforms.
- Deployed to **700+ users, saving 1,000+ manual hours annually**; now a business-critical system.
- Drove organization-wide adoption, scaling **platform usage from 1K to over 20K monthly** active users.
- Proficient in **React, TypeScript, FastAPI, AWS (Lambda, CDK, S3, DynamoDB), Docker, and Azure OpenAI**, with a strong foundation in building serverless microservices and automated pipelines.
- Proven track record in process automation, OCR pipelines, and data-driven financial applications, delivering measurable efficiency gains and operational impact.

Work Experience

Koch Industries, Bangalore **Mar'2023-Present**
Software Engineer – Data & Analytics COE

Key Result Areas:

FAM CIP (Fixed Assets Management – Construction In Progress)

- Designed and led the end-to-end development of a workflow automation platform streamlining finance operations.
- Conducted stakeholder discovery and mapped legacy finance processes for digital transformation.
- Designed UI/UX in Figma and implemented the front-end in React with a Serverless back-end.
- Built secure SSO and RBAC authentication, deployed across UAT, Non-Prod, and Prod environments.
- Engineered automated email pipelines using AWS Lambda and SES.

Koch GPT – Internal Generative AI Platform

- Revamped the platform UI using a custom Material UI design system, leading to a 30% rise in user engagement.
- Led product design and UX strategy, ensuring consistent user experience across modules.
- Designed a plugin onboarding microservice framework to scale access for external teams.
- Designed back-end architecture for chat history, prompt templates, and plugin configurations.

File-Ops OCR Service

- Built a modular OCR pipeline integrating Python Fitz, AWS Textract, and Azure OpenAI (GPT-4o) for intelligent document parsing.
- Designed adaptive logic to switch between vision-based and native parsing based on PDF metadata.
- Implemented multi-worker concurrency for high-throughput page processing.
- Packaged as a Dockerized microservice, deployed via AWS Lambda & API Gateway, and integrated into internal REST APIs.
- Adopted across KochGPT and multiple automation workflow.

GLAD (General Ledger Anomaly Detection)

- Reduced manual reporting time for finance controllers by 10+ hours/month via an interactive variance analysis dashboard.
- Built a robust data processing layer managing 500K+ transactions/month for YOY trend analysis and anoPmaly detection.

Education

- **B.E. in Computer Science**, CMR Institute of Technology, Bangalore, 2023, GPA: 8.1

Contact

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Core Competencies

- Software Development
- Full-stack Development
- Requirement Gathering
- Incident Resolution
- Workflow Automation
- Root Cause Analysis
- Scrum Frameworks
- Agile Methodology

Technical Skills

- **Languages**: JavaScript, TypeScript, Python, SQL (Basic)
- **Front-end**: React, Redux, Figma (UI/UX Design), TailwindCSS, Material UI, Shadcn/ui
- **Back-end & APIs:** FastAPI, REST APIs, JWT Auth, SSO (SAML), Microservices
- **Cloud & DevOps**: AWS (Lambda, CDK, S3, DynamoDB, SES, Textract), Azure OpenAI, LLMs, GitLab CI/CD, Docker
- **Tooling**: Git, Postman, Webflow, Serverless Framework
- **Core Concepts**: System Design, OCR Pipelines, Prompt Engineering, Scalable Architecture, Generative AI

Part-time Experience

KPI Media-Remote, Singapore
Designer & Developer | Mar'21-Jan'22

Key Result Area:

- Redesigned and developed marketing websites on Webflow, integrating custom code to enhance functionality and performance.
- Led a complete revamp of the company's site, kpimedia.sg, resulting in a 40% increase in web traffic through improved design and SEO optimization.

Personal Details

- **Date of Birth**: 13th July 2000
- **Languages**: English, Hindi
- **Address**: Bangalore, 560037